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MILITARY SPECIFICATION

MILLING MACHINES, BED TYPE, VERTICAL, DOUBLE HOUSING, NUMERICALLY CONTROLLED, PROFILING, 5 TO 25 HORSEPOWER PER SPINDLE, SINGLE AND MULTI-SPINDLE MACHINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, bed type, double housing vertical profilers, with horsepower range from 5 to 25, table widths from 30 to 120 inches and table lengths up to 312 inches, single and multiple motor-driven spindles, with equipment and accessories as described herein.

1.2 Classification. Milling machines covered by this specification are of the following types, classes and sizes. The type, class and size furnished shall be as specified (see 6.2).

Type I - Single Spindle Machine.

Type II - Multi-spindle Machine - each spindle assembly constructed as an individual unit and driven independently.

Type III - Multi-spindle Machine - two or more spindles mounted in a single housing and driven by one motor through a common spindle drive.

Type I machines shall include Class 1 only. Type II and Type III machines shall include Classes 2 through 4.

Class 1 - One spindle machine
Class 2 - Two spindle machine
Class 3 - Three spindle machine
Class 4 - Four spindle machine

FSC 3417

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Classes 1 and 2 shall include Sizes 1 through 10. Classes 3 and 4 shall include Sizes 3 through 10 only.

Size 1 - 30 inches minimum table width
Size 2 - 36 inches minimum table width
Size 3 - 48 inches minimum table width
Size 4 - 60 inches minimum table width
Size 5 - 66 inches minimum table width
Size 6 - 72 inches minimum table width
Size 7 - 78 inches minimum table width
Size 8 - 84 inches minimum table width
Size 9 - 96 inches minimum table width
Size 10 - 120 inches minimum table width

2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NATIONAL MACHINE TOOL BUILDERS' ASSOCIATION

Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA) STANDARDS

MG1 Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th St., New York, NY 10017.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360.

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., N. W., Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS

ANS B5.1 T-Slots - Their Bolts, Nuts, Tongues and Cutters

ANS B5.18 Spindle Noses and Arbors for Milling Machines

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

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ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-227	One Inch Perforated Paper Tape
RS-244	Character Codes for Numerical Machine Control Perforated Tape
RS-267	Axis and Motion Nomenclature for Numerically Controlled Machines
RS-274	Interchangeable Perforated Tape Variable Block Format for Contouring and Contouring/Positioning Numerically Controlled Machines
RS-281	Construction Standards-Numerical Machine Tool Control
RS-358	Subset of USA Standard Code for Information Interchange for Numerical Machine Control Perforated Tape

(Application for copies should be addressed to Electronic Industries Association, Engineering Dept., 2001 Eye St., N. W., Washington, DC 20006.)

NATIONAL AEROSPACE INDUSTRIES ASSOCIATION STANDARD NAS 979

Uniform Cutting Tests

(Application for copies should be addressed to Aerospace Industries Association, 1725 De Sales St., N. W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First Article approval. If First Article approval is required (see 6.2 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

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3.2 Design. The machine and control system shall be new and one of the manufacturer's current models. The machine shall have a floor-mounted base for the purpose of supporting the table and a horizontally mounted rail supported on two floor-mounted housings vertically fixed on both sides of the bed. The spindle(s) shall be mounted vertically on the horizontally mounted bridge rail in line across the table. The machine shall be capable of automatically performing milling and profiling operations by controlling the axes of motion individually, in any combination of two axes, and all three axes simultaneously. The machine axes of movement shall be: (1) longitudinal (X) - table moving along the floor-mounted base; (2) cross (Y) - the saddle moving along the horizontally mounted bridge rail; (3) vertical (Z) - the spindle headstock moving vertically along the saddle. All numerically controlled axes of motion shall be capable of both automatic control by programmed numerical data media and manual control. The machine and control system design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily and easily accessible for adjustment, replacement, and repair.

3.2.1 Operating controls. All operating controls necessary for manual and tape operation of the machine shall be mounted on a control console or pendant and clearly identified to facilitate use. Voltage and current meters, digital readouts, indicator lights and all operational control aids listed herein shall be arranged and mounted in full view of the operator. An emergency stop switch shall be furnished on each fixed control station and on each portable and remote control station.

* 3.2.2 Safety measures. The machine and control unit shall be equipped with safety measures for protection of the operator and equipment. Safety devices such as limit trips, positive stops and warning signals, shall protect the equipment against overload, overtravel, and malfunction of a component. High voltage power supply cabinet doors shall be provided with electrical interlocks. The operator console shall be equipped with a key lock type master switch which shall deactivate the control and power unit to prevent starting of the machine by unauthorized persons. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

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* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. An automatic lubricating system shall deliver lubricating oil to all mating gears, sliding parts, and bearings except sealed-for-life type. Each recirculating system shall include a filter which can be cleaned or replaced. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible. The lubrication system shall have a low pressure warning system to alert the operator and stop the spindle and axis movement under low pressure conditions.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by the manufacturer.

* 3.3 Construction. The machine shall be so constructed that when installed and connected to power, it will be ready for operation. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale and mismatching. No process such as welding, peening, plugging or filling with solder or paste shall be used for reclaiming any defective part.

* 3.3.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where required in the original design. None of these operations shall be employed as a repair measure for any defective part.

* 3.3.3 Fastening devices. All screws, pins, bolts, and similar internal and external parts shall be installed with means for preventing change of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

* 3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, sprues, fins, flash, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Dials. All feed screw dials shall be graduated in the inch system. Graduations to indicate stock removal or tool movement shall be in increments of not more than 0.001 inch. Dial diameter shall be such that the graduations may be easily read. Dial type mode switches shall be marked by appropriate methods to clearly indicate type function to be performed. Degree dials and scales shall be graduated in not more than one-degree increments and stamped or etched with numerals to show each not more than five-degree increment throughout its range. Dials shall be permanently and legibly engraved or etched and shall have a non-glare finish.

* 3.3.8 Readouts. All digital readouts shall display data in illuminated figures at least 1/2 inch high that give direct readings without requiring any calculation or interpolation. Each digit of the readout display shall be of the in-line type.

* 3.3.9 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, permanently and legibly displayed in bold-faced letters on a contrasting background.

* 3.3.10 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

3.4 Components. The milling machine and control system furnished under this specification shall have components as described below.

* 3.4.1 Bed. The bed shall be a casting of iron alloy or fabricated from steel. The bed shall contain the feed drive mechanism for the table and have the necessary access doors or covered openings to facilitate required adjustments and replacement of parts. The length of the bed and bed ways shall be such that the worktable is fully supported and does not overhang the bed in any position within its rated travel. An adequate number of leveling screws and holddown bolts shall be provided to enable the machine to be properly installed and leveled.

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* 3.4.2 Table. The table shall be a casting of iron, iron alloy or steel. The table shall be supported on anti-friction roller bearings on replaceable way strips fitted on the bedways with means to adjust the ways. The location of the table on the bed shall be such that the table will be fully supported throughout its range of travel. Unless otherwise specified (see 6.2), the table shall have not less than one T-slot conforming to ANS B5.1 every five inches, parallel to the longitudinal travel of the table. The table shall have coolant drain channels for returning the coolant to the reservoir with provision to prevent chips from entering the reservoir. Means shall be provided to prevent overtravel of the table. The table shall be provided with power and rapid traverse for longitudinal movement.

* 3.4.3 Housings. Each housing shall be a casting of iron or iron alloy, or a steel weldment. The housings shall adequately support the bridge rail and spindle assembly. The tops of the housings shall have provisions for mounting the bridge rail. The housings shall be provided with access doors as required for servicing the components located therein. The bottom of each housing shall be provided with an adequate number of holddown bolts as well as leveling adjustment screws located to assure and maintain level of machine under maximum load conditions.

* 3.4.4 Bridge rail. The bridge rail shall be designed and constructed for rigidity, stability and minimum sag or deflection. The bridge shall be mounted on the top of the vertical housings parallel to the table surface. The bridge shall have hardened and ground ways of large proportions to provide the spindle and saddle with maximum support.

* 3.4.5 Spindle(s) saddle. The saddle, for the purposes of this specification, is defined as the structural part of the machine that moves along the ways of the fixed bridge rail parallel to the table surface and provides ways for the spindle(s) headstock to travel in a plane perpendicular to the table surface. The saddle shall be constructed for strength and rigidity to assure accurate vibration-free performance within the full range and capacity of the machine.

* 3.4.6 Spindle headstock. The headstock, carrying one or more spindles, shall be a vertical sliding structure mounted on the saddle ways, adequately proportioned to provide sufficient rigidity of the machine. The headstock shall be supported to the saddle by way strips or anti-friction roller bearings. Simple means of adjustment to insure correct alignment of the mating parts shall be furnished.

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* 3.4.7 Spindle assembly. Each spindle shall be made of heat-treated alloy steel. When multi-spindles are required, the spindles shall be identical and shall be mounted to the headstock in line parallel to the table working surface. Spindle bearings shall be designed and assembled to compensate for thermal expansion. Means for maintaining a controlled temperature in the headstock spindle shall be provided. The spindle shall have precision ground Number 50 American National Standard Taper and shall be in accordance with ANS B5.18. Spindle(s) shall have forward and reverse directions of rotation, controllable by electro-manual controls and the NC unit. The spindle(s) shall have features to alert the operator and cause all movements to stop if any spindle becomes overloaded, stalled, or overheated. Braking devices shall be interlocked to prevent power rotation of spindle(s) until the brake device has been released. Means shall be provided for selection of spindle speeds by electro-manual controls and, when specified (see 6.2), by tape program. When dual horsepower spindle motors are required (see 6.2), the lowest horsepower shall not be less than that stated in Table I.

* 3.4.8 Feed drive. The X, Y, and Z axes of the machine shall be actuated through a closed loop, high gain electric or electro-hydraulic servomechanism driving recirculating anti-backlash ball screws and pre-loaded recirculating ball screw nuts or rack and pinion drives. Infinitely variable feed drives shall be provided for all axes, actuated and controlled by commands from the control tape and by manually inserting data.

* 3.4.9 Power rapid traverse. Power rapid traverse shall be provided for movement of X, Y, and Z slides. The rate of rapid traverse shall be independent of any feed adjustment.

* 3.4.10 Way wipers and covers. X and Y axes ways shall be provided with wipers and covers to protect the way bearing surfaces from dust and other foreign matter. Z ways shall be furnished with wipers or covers.

* 3.4.11 Machine control unit (MCU). Unless otherwise specified (see 6.2), the MCU shall be as required in 3.4.11 through 3.4.11.26. The control unit shall be in accordance with the applicable requirements of EIA RS-281 and the JIC Electrical Standard for General Purpose Machine Tools. The MCU shall be a three-axis contouring control system capable of accepting program data and translating this data into motion command signals as well as initiating the programmed auxiliary functions. The system shall be capable of controlling the axes of motion individually, any combination of two axes, and all three axes simultaneously. Axes designations shall be in accordance with EIA RS-267. All systems shall have linear interpolation. Input shall

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be in the form of binary coded decimal commands introduced by perforated tape and manual data input. The logic circuitry of the control shall be solid state design (discrete solid state components, integrated circuits or a combination) of modular construction. Plug-in circuit boards shall be keyed and identified by part number to facilitate location in the system. The system shall be capable of operating in an ambient temperature of 60°F. to 110°F. with a humidity of 40 to 95 percent and shall function normally at 100 percent efficiency with line voltage fluctuation of ± 10 percent. The control system shall be complete with reels and tape feed mechanism, and necessary control circuitry and other accessories specified herein, properly installed within the control cabinet. The control system shall have the following standard features:

Manual data input	3.4.11.1	Block delete	3.4.11.14
Tape data input	3.4.11.2	Parity check	3.4.11.15
Preparatory function	3.4.11.3	Synchronization	3.4.11.16
Miscellaneous function	3.4.11.4	Zero synchronization	3.4.11.17
Interpolation	3.4.11.5	Full floating zero	3.4.11.18
Selectable input character codes	3.4.11.6	Position feedback	3.4.11.19
Programmed stop	3.4.11.7	Storage switch	3.4.11.20
Input resolution	3.4.11.8	Data storage	3.4.11.21
Operation mode selection	3.4.11.9	Mirror image	3.4.11.22
Departure control	3.4.11.10	Acceleration & deceleration	3.4.11.23
Tape reader	3.4.11.11	Feed rate override	3.4.11.24
Sprocket hole verification	3.4.11.12	Postprocessor	3.4.11.25
Sequence number readout	3.4.11.13	Equipment cabinet	3.4.11.26

3.4.11.1 Manual data input. Manual data input capability shall be provided in the MCU and shall provide a method for inserting manually one block of data to accomplish any operation within the capabilities of the machine under linear mode of operation. Appropriate dials and operator controls shall be provided on the operator console with necessary identification of dials and controls.

3.4.11.2 Tape data input. Input shall be from 1-inch wide, 8-channel, perforated tape in accordance with EIA RS-227. Tape format shall be variable block, word address in accordance with EIA RS-274. The least significant digit of the incremental dimension data shall be not more than 0.0001 inch.

* 3.4.11.3 Preparatory function. All preparatory functions shall be coded in accordance with EIA RS-274, Appendix B. Preparatory functions shall be furnished as specified (see 6.2), and unless otherwise specified, the machine builder shall use unassigned preparatory codes as needed. The unassigned preparatory function codes shall not be used in conflict with preassigned functions.

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* 3.4.11.4 Miscellaneous function. All miscellaneous functions shall be codes in accordance with EIA RS-274, Appendix B. Miscellaneous functions shall be furnished as specified (see 6.2), and unless otherwise specified, the machine builder shall use unassigned miscellaneous function codes as needed. The unassigned miscellaneous functions shall not be used in conflict with pre-assigned functions. Miscellaneous function codes shall include, but shall not be limited to, the following:

Program Stop	Spindle CW	Coolant Off
Optional Stop	Spindle CCW	Coolant On
End of Program	Spindle Off	End of Tape

* 3.4.11.5 Interpolation. The interpolation function shall be an integral part of the machine control unit. The basic control system shall perform contouring cuts by simultaneous incremental or absolute movements of any or all axes. When specified under 3.12, circular or parabolic interpolation shall be furnished.

* 3.4.11.6 Selectable input character codes. The control unit shall have the capability of receiving and acting on instructions coded in accordance with EIA RS-244 and RS-358. Means shall be provided for the control to accept the code utilized in the control tape.

* 3.4.11.7 Programmed stop. The system shall be such that stops can be programmed as desired for such purposes as cutter change, cutter and part inspection, spindle speed change, and program checks. Resumption of automatic operation shall not cause any loss of accuracy on the workpiece. A monitor light shall signal a programmed stop.

* 3.4.11.8 Input resolution. The input resolution of the machine shall not be more than 0.0001 inch.

* 3.4.11.9 Operation mode selection. Mode selection switch shall be provided to govern the mode of operation in which the control is to operate. The selector switch shall enable the operator to switch from tape control to a manual mode for insertion of data by means of the manual-data input control. The manual-data input mode shall permit the operator to duplicate tape operation by manually entering data into the control by manual-data input selector switches provided for that purpose. The mode selector switch shall be located on operator's control panel.

* 3.4.11.10 Departure control. The departure commands shall command movement in distance and direction of all axes. The maximum departure by single command for any mode of departure shall be not less than 99.999 inches. The smallest programmed increment of movement of the slides shall be not more than 0.0001 inch.

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* 3.4.11.11 Tape reader. The tape reader shall be capable of feeding and reading 1-inch, 8-channel, perforated tape with characteristics required by EIA RS-227. The reader shall be enclosed in a dust-free sealed cabinet and shall be visible and accessible for removal, replacement, and maintenance of tape. The tape reader shall be solid state photoelectric capable of reading a minimum of 300 characters per second with power input frequency of 60 Hz. The tape reader shall have both manual and automatic modes of operation, including rewinding of tape.

* 3.4.11.12 Sprocket hole verification. A sprocket hole verification circuit shall be provided. Absence or misalignment of the sprocket holes shall cause a halt in machine function and illumination of a reading error light on the operator's control panel.

* 3.4.11.13 Sequence number readout. There shall be incorporated into the machine control system an operation sequence number readout having at least three digits.

* 3.4.11.14 Block delete. The control shall have means for by-passing blocks of tape when any properly coded block of the programmed cycle is not desired for certain machine operations.

* 3.4.11.15 Parity check. The machine control unit shall have a parity check system. The electronic circuitry of the system shall have means to stop all functions of the machine when an improper number of holes is read in a transverse row of tape. The operator's control panel shall incorporate a parity error lamp illuminated by parity error.

* 3.4.11.16 Synchronization. Any loss in tape-to-part synchronization shall automatically stop operation. The system shall have circuits to monitor the loss of synchronization which shall be indicated by an error light. Error light shall be located in a position that can be seen from the operator's normal work station. The synchronization system shall be designed to make it possible for the operator to stop the machine, withdraw the spindle, change cutters, return cutter to machining position and resume machining under tape control without loss of synchronization.

* 3.4.11.17 Zero synchronization. The machine control shall have the ability to synchronize the machine slides automatically to the original zero set point, when manually positioned within 0.04 inch of the original set point.

* 3.4.11.18 Full floating zero. The control system shall have means for locating the zero reference set point of each axis at any location within the work area.

* 3.4.11.19 Position feedback. The feedback system shall be a closed loop system to correspond command data and slide positions.

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* 3.4.11.20 Storage switch. The MCU shall employ controls for clearing temporary and active storage. The control shall be located on the operator's control panel.

* 3.4.11.21 Data storage. The machine shall have a buffer storage unit or other means for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the function. The internal storage shall delay or store data until allocated by the controlling devices.

* 3.4.11.22 Mirror image. The machine shall be capable of producing either right or left-hand parts from the same tape through the use of inversion switches for each axis. Inversion switches for each axis shall be located on the operator's control cabinet. The system shall have indicating lights showing reverse settings and shall be visible to the operator. Mirror image shall not affect the direction sense of manual controls.

* 3.4.11.23 Acceleration and deceleration. The MCU shall include an acceleration and deceleration function which shall not require programming into the input data.

* 3.4.11.24 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from 100 percent to 10 percent of the programmed rate in continuous adjustment or in not less than 10 steps. The feed rate override control shall have an OFF mode that will halt all motion. The feed rate control shall be mounted in a location convenient to the operator.

* 3.4.11.25 Postprocessor. A postprocessor having the specified additional characteristics shall be furnished with the numerical control unit to convert control data generated by a specified computer system (see 6.2), to a format acceptable to the numerical control unit. The postprocessor shall consist of computer input in the form of cards or tape and related instructions acceptable to the computer. The postprocessor input supplementing the key punch output of the program manuscript shall enable the computer to convert all information for a parts program into programmed instructions acceptable to the machine and its controls, and produce parts within the tolerance specified for the machine.

* 3.4.11.26 Equipment cabinet. The MCU cabinet shall contain the electronic power supply, tape readers, all associated wiring and components for the numerical control system. The enclosure shall be constructed to minimize entrance of dust, moisture and other environmental contaminants that may cause damage to the system. Components sensitive to vibration shall be protected from vibration. The enclosure shall be of such construction as to provide for easy access to circuits for maintenance and shall conform to JIC Electrical Standards or EIA RS-281 as applicable.

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* 3.4.12 Maintenance and test features. Maintenance and test features shall be furnished including operational test circuitry which shall provide a diagnostic means for locating malfunctions of electronic components. Test points shall be provided at convenient locations for measurement and adjustment purposes and for the observation of critical waveforms which denote proper functioning of circuitry. Operating voltages shall be indicated at all important test points. Electronic and electrical components shall be clearly marked and identified for ease of maintenance. Test circuits shall be included to permit the detection of marginal or decaying electronic components that might fail under normal operating conditions.

* 3.5 Sizes and capacities. Milling machines covered by this specification shall meet the requirements of Table I. Options shall be furnished only when specified.

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TABLE I

CHARACTERISTICS	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10
Table size, working surface										
Width, inches	30	36	48	60	66	72	78	84	96	120
Length, inches	36	48	48	72	96	96	96	96	120	144
Optional lengths	48, 60	60, 72	72, 96, 120	96, 120, 144	120, 144	120, 144	120, 144	144, 168, 192, 214	144, 168, 192, 214	192, 214, 240, 312
Feed range, inches										
Longitudinal (X)	NOT LESS THAN TABLE LENGTH									
Cross (Y) single spindle	AT LEAST 4 INCHES MORE THAN TABLE WIDTH									
Cross (Y) multi-spindle	NOT LESS THAN DISTANCE BETWEEN SPINDLES									
Vertical (Z)	12	14	14	14	18	18	18	18	18	18
Quill movement or depth adjustment	1	1	1	1	1	1	4	4	4	4
Spindle nose to table, inches	18	18	18	18	18	18	18	18	18	18
Feed rates, number of	INFINITE									
Feed rate, range, IPM, NC										
Longitudinal (X)	½ to 100	½ to 100	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150
Cross (Y)	½ to 100	½ to 100	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150
Vertical (Z)	½ to 100	½ to 100	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150	0 to 150
Feed rate, range, IPM, manual										
Longitudinal (X)	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100
Cross (Y)	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100	0 to 100
Vertical (Z)	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60	0 to 60
Power rapid positioning, IPM										
Longitudinal (X)	100	100	150	150	150	150	150	150	150	150
Cross (Y)	100	100	150	150	150	150	150	150	150	150
Vertical (Z)	100	100	150	150	150	150	150	150	150	150
T-slots size	13/16	13/16	13/16	13/16	13/16	13/16	13/16	13/16	13/16	13/16
T-slots center to center distance, inches	5	5	5	5	5	5	5	5	5	5
Number of spindles	1	1	1	1	1	1	1	1	1	1
Optional number of spindles	2	2	2 or 3	2, 3 or 4	2, 3 or 4	2, 3 or 4	2, 3 or 4	2, 3 or 4	2, 3 or 4	2, 3 or 4
Distance between spindles										
2 spindles	12	15	20	30	33	36	38	42	48	60
3 spindles	-	-	15	18	20	22	24	30	30	30
4 spindles	-	-	-	15	16	18	19	21	24	30
Spindle motor, horsepower per spindle	5	5	5	5	5	5	5	5	5	5
Optional horsepower per spindle	7½, 10, 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25	7½, 10 15, 20 or 25
Spindle speeds, number of	INFINITE									
Optional number of speeds	24	24	24	24	24	24	24	24	24	24
Spindle speed range, RPM	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600	36 to 3600
Optional range	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800	24 to 2400 or 18 to 1800
Spindle horsepower per RPM	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Optional HP/RPM	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10
Distance between housings, inches	42	48	60	72	78	84	90	96	108	132

Requirements are not less than those shown. Where a range is shown, the performance shall encompass the range from the stated minimum or less to the stated maximu or greater.

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3.6 Performance. Each machine shall be capable of satisfactory performance when tested in accordance with 4.5.3 through 4.5.3.4.

* 3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of Table II.

TABLE II

TEST	TOLERANCES
Spindle(s) periphery runout	0.0004 inch TIR
Spindle(s) face runout	0.0003 inch TIR
Spindle(s) internal taper runout	
1 inch from spindle(s) nose	0.0004 inch
12 inches from spindle(s) nose	0.001 inch
Longitudinal rise and fall of table	0.0005 inch in any 12-inch span Total not to exceed 0.0005 + 0.00015 per 12 inches of travel in excess of 12 inches
Crossrail ways parallel with table surface	0.0005 inch in any 12-inch span Total not to exceed 0.0005 + 0.00015 per 12 inches of travel in excess of 12 inches
Spindle(s) square with table (center of rotation)	0.001 per foot
Spindle(s) travel square with table	0.0005 inch in any 12-inch span Total not to exceed 0.0005 + 0.00015 per 12 inches of travel in excess of 12 inches
T-Slots parallel to table travel	0.0005 inch in any 12-inch span Total not to exceed 0.0005 + 0.00015 per 12 inches of travel in excess of 12 inches
Spindle centerlines with respect to each other	± 0.002 inch

* 3.8 Electrical equipment. Unless otherwise specified (see 6.2), the electrical system shall conform to JIC Electrical Standards for General Purpose Machine Tools or RS-281 as applicable. Unless otherwise specified (see 6.2), the machine shall operate from 230/460 volt, 3 phase, 60 Hz power supply,

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initially connected for operation on the voltage specified. An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor in Table 15-1 of JIC of Electrical Standards.

* 3.8.1 Motors. The electric spindle drive motors shall be single or variable speed. Motors shall be either AC, 3 phase, 60 Hz, or industrial DC conforming to NEMA Standard MG1. Rectifiers for DC motors shall be solid state. Unless otherwise specified (see 6.2), each motor shall meet the requirements of a drip-proof enclosure.

* 3.9 Coolant system. A coolant system shall be furnished, including a sump or reservoir, a power-driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit full flow of coolant, and shall have means for draining and cleaning. The system shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant flow. When required (see 6.2), a heat exchanger for coolant shall be furnished.

* 3.10 Hydraulic system. A hydraulic system shall be furnished complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. The hydraulic system shall be provided with a heat exchanger of capacity adequate to hold the temperature of the hydraulic fluid within limits required for satisfactory operation for hydraulically controlled machine components in ambient temperatures up to 125°F. The system shall be provided with thermal switches and signalling devices to alert the operator to existence of excessive temperature in the hydraulic system and to shut down the system when safe maximum limits are reached.

* 3.11 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. Electrical load indicator(s). Each spindle drive motor shall have an electrical load indicator. The dial shall indicate either amperes or percent of rated full load current, and shall be clearly marked to indicate the rated current for continuous operation of the machine. The accuracy of the meter shall be within two percent at the full rated load current of the motor. The instrument shall be clearly readable from the operator's normal work position.

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c. Elapsed time meter. The control system shall be provided with an elapsed time meter of the non-reset type which shall be capable of indicating accumulated operating time to at least 9,999.9 hours.

* 3.12 Optional machine and machine control equipment. When required, the following control features shall be furnished in the numerical control system as specified (see 6.2):

Mist coolant	3.12.1	Parabolic interpolation	3.12.9
Chip removal system	3.12.2	Switchable position readout	3.12.10
Power drawbolts	3.12.3	Individual position readouts	3.12.11
Cutter compensation	3.12.4	High speed, photoelectric tape reader	3.12.12
Tape reader overshoot	3.12.5	Dual tape readers	3.12.13
Block-by-block read-in	3.12.6	Feed rate number	3.12.14
Sequence number search and display	3.12.7	Air conditioning	3.12.15
Circular interpolation	3.12.8		

* 3.12.1 Mist coolant. When specified, the machine shall be furnished with mist coolant system in addition to the system required in 3.9. The system shall be adequate for all types of machining and shall be provided with means for controlling the flow of coolant. All features necessary for draining and cleaning, including troughs, drain plugs, pumps, piping, control nozzles and sump shall be furnished. Coolant on and off, and the selection of the type coolant system shall be controlled by command from the tape and electro-manual control.

* 3.12.2 Chip removal system. When specified, a chip removal system shall be furnished to transport chips to a collection area.

* 3.12.3 Power drawbolts. When specified, the spindle(s) shall be furnished with power drawbolts. The power drawbolt mechanism shall have means to prevent spindle rotation during actuation of the power drawbolts.

* 3.12.4 Cutter compensation. When specified, means shall be provided to compensate for cutter diameter differing from that utilized in programming. The cutter compensation shall have a range of at least ± 0.100 inch on the diameter.

* 3.12.5 Tape reader overshoot. When specified, the machine control unit shall incorporate a tape reader overshoot detection circuit that shall stop the machine, should the tape overshoot the end of block code.

* 3.12.6 Block-by-block read-in. When specified, the machine control shall be provided with a block-by-block read-in function. Block-by-block read-in

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function shall provide the operator means of reading information into the control one block at a time.

* 3.12.7 Sequence number search and display. When specified, the machine shall be provided with a capability of advancing or retracting tape manually by operator control and with an automatic control which causes the tape to move to a desired sequence number. In both modes the visual display shall continuously indicate the sequence number passing through the reader. Appropriate control switches or push buttons shall be provided and clearly identified.

* 3.12.8 Circular interpolation. When specified, circular interpolation shall be furnished to provide a means within the control system for generating an arc in any one quadrant using one block of commands on the tape. Direction of the tool-center arc shall be clockwise or counterclockwise and shall be selected by a preparatory function programmed into the tape. The least significant digit of dimensional data shall be not more than 0.001 inch.

* 3.12.9 Parabolic interpolation. When specified, parabolic interpolation shall be furnished. The parabolic interpolation system shall have the ability to generate the link between programmed points and smooth curving of three dimensional shapes in a plane of the machine slides.

* 3.12.10 Switchable position readout. When specified, the control system shall be supplied with switchable axis position readout. The switchable position readout shall be capable of displaying full travel of each axis in integers and decimal values equivalent to the resolution for each axis. The readouts shall be command data, or absolute position, as specified.

* 3.12.11 Individual position readouts. When specified, the control system shall be furnished with individual axis position readouts. The position readouts shall be capable of readings displaying full travel in integers and decimal values equivalent to the resolution for each axis. The positions shall be displayed at all times when the machine is powered. The readouts shall be command data, or absolute position, as specified.

* 3.12.12 High speed, photoelectric tape reader. When specified, a tape reader shall be furnished capable of feeding and reading a minimum of 500 characters per second of 1-inch, 8-channel perforated tape. The tape reader shall be solid state photoelectric type. Tape reels shall hold at least 1200 feet of 0.0026 inch thick mylar tape. The tape reader shall have both manual and automatic modes of operation including rewinding of tape.

* 3.12.13 Dual tape readers. When specified, an additional tape reader shall be furnished. The dual reader system shall provide rapid program switching at the end of the cycle. The readers shall be in accordance with paragraph 3.4.11.11 unless otherwise specified.

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* 3.12.14 Feed rate number. When specified, velocity/distance calculation, 3-digit feed rate, or switchable feed rate numbers shall be furnished with the numerical control unit.

* 3.12.15 Air conditioning. When specified, the control cabinet shall be equipped with an air conditioning unit capable of maintaining temperature and humidity within the cabinet so that the unit will operate efficiently in ambient temperature ranging from 60° to 125° with relative humidity of 20 to 95 percent.

* 3.13 Additional accessories and equipment. Additional accessories, equipment, or attachments shall be furnished as specified (see 6.2).

* 3.14 Electromagnetic interference control. Unless otherwise specified (see 6.2), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

* 3.15 Fungus control. When required (see 6.2), fungus proofing shall be as specified.

3.16 Lubrication chart or plate. Unless otherwise specified (see 6.2), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Federal or Military Specification Number (if available)

3.17 Nameplate. Unless otherwise specified (see 6.2), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity:

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Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

* 3.18 Identification plate. When specified (see 6.2), a separate identification plate of corrosion-resistant metal permanently and legibly marked in letters not less than 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely attached to each machine. The plate shall have blank space for the government to stamp 25 characters of the same size as the caption.

* 3.19 Control element identification. Unless otherwise specified (see 6.2), each separately housed part of the numerical control shall have attached to the housing a plate identifying the control as follows:

Control manufacturer _____

Control Model No. _____

Control Serial No. _____

Initially sold for use with machine:

Manufacturer _____

Model No. _____

Serial No. _____

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The required information may be given in any suitable order or format that will identify the control and show its historical connection with the machine. The manufacturer or control builder shall maintain a permanent record of all basic data pertaining to the numerical control furnished with the machine including tape format, tape reader, interpolation capabilities, postprocessor, and the computer or computers compatible with the control and its postprocessor. Data pertaining to postprocessor shall be retained in sufficient detail on punched cards, magnetic tape or other suitable media to permit reproduction, if required. The manufacturer shall furnish a brief outline statement of the control features and capabilities, preparatory and miscellaneous codes, tape format, and pertinent postprocessor and computer information. This information shall be on a printed sheet securely attached to the inner surface of the control cabinet. The abbreviated format specified by EIA RS-274 may be used for this purpose, in lieu of printed outline statement.

* 3.20 Technical data. Technical data shall be furnished as specified (see 6.2.2).

* 3.21 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the First Article.

* 4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.3.4, and the inspection in 4.6. Failure of an item to pass any examination or test shall be cause for rejection.

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* 4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.21.

* 4.5 Test. The machine and its numerical control system shall be subjected to the following tests. All instruments required to perform and evaluate tests shall have evidence of calibration traceable to the National Bureau of Standards. The test tapes and computer printouts shall become the property of the government.

* 4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes for an initial warm-up period. Proper operation of all manual controls, motors, adjusting mechanisms, and accessories shall be verified during the warm-up period. Upon completion of the initial check procedure, the system shall be cycled continually for three hours under tape control to demonstrate operability of the equipment and permit the machine to reach a steady operating temperature. The test shall insure that the reader decoding unit and other systems are properly functioning and present correct machine control information to the machine tool. The program shall also include rapid travel of the slides, independently and simultaneously. A test shall be performed of parity check to demonstrate that the machine will stop when parity is not obtained due to improper code punching or reading error. If a malfunction occurs during the cycling period, the tape shall be restarted each time until a full 3-hour programmed cycle is completed without a malfunction. Spindle speed and milling feed rates shall be verified.

* 4.5.2 Alignment accuracy test. The machine shall be tested to determine compliance with the alignment accuracy requirements of 3.7. The measuring equipment and machine shall have reached a steady temperature before making any test readings.

* 4.5.3 Performance tests. The following tests shall be performed to test the functional accuracy, dynamic stiffness, and servo response of the machine and control under operating conditions.

* 4.5.3.1 Metal removal and maximum horsepower test. The material for this test shall be SAE/AISI 4140 or 4340 annealed steel with Brinell hardness in the range 300 to 350. Not less than four workpieces shall be used, having a thickness not less than 2.5 inches, and an area of the largest surface not less than 144 square inches. The workpieces shall be placed on the worktable to cover a range requiring not less than 75 percent of the cross and longitudinal travels. The spindle speed and feed rates shall be such that the chip load is not less than 0.012 and the peripheral cutter speed not more than 325 surface feed per minute. The depth of cut shall be sufficient

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to load the spindle motor to not less than its maximum rated input horsepower. The load meter specified in 3.11.b is acceptable for measuring spindle motor load, or other instruments may be used. Except as limited herein, the cutter speed, feed rate, cutter type and cutter diameter shall be selected by the supplier. The cutting action shall be smooth and even. The surface of the workpieces after the cut shall show no visual evidence of tool chatter and the surface of the workpieces shall meet the following requirements:

Finish	150 RHR
Flatness	0.002" per 1-ft. span, plus 0.0002" per additional ft.

Metal removal shall be not less than nameplate HP x 0.65 cu. in. per minute. All spindles of multi-spindle machines shall be tested.

* 4.5.3.2 Overshoot and undercut test. The overshoot and undercut test shall be performed in accordance with 4.3.3.4.1 of NAS 979, Sheet 19, Copyright 1969, to test the dynamic servo response of the machine and control. Tolerances for the completed test piece shall not exceed the following:

Overshoot and undercut in X and Y axes	0.0015 for Test A
Overshoot and undercut in X and Y axes	0.002 for Test B
Overshoot in Z axis	0.0015 for Test A
Overshoot in Z axis	0.002 for Test B

* 4.5.3.3 Composite cutting test. The composite cutting test shall be performed in accordance with paragraph 4.3.3.5.1 of NAS 979, Sheets 23 and 24, Copyright 1969, to test the profiling and contouring capabilities of the machine and control unit. A feed rate of 25 IPM and a spindle speed of approximately 1600 RPM shall be used for this test. Tolerances for the completed test piece shall be as follows:

Squareness	0.002" TIR (outside square, canted square)
Parallelism	0.002" TIR (outside square, canted square)
Angularity of ramp	0.0025" TIR (5-degree ramp)
Circle	+0.0025" on diameter
Flatness	0.002" TIR (flat surface of outside square)
Finish	35 RHR, all surfaces

* 4.5.3.4 Positioning accuracy and repeatability. The positioning accuracy and repeatability of the machine slides shall be evaluated in accordance with the NMTBA Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools, Section 1 - Linear Definition. At least ten random target points, spaced over the full travel of each axis shall be selected and approached from both directions.

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* 4.5.4 Electromagnetic interference control tests. Unless otherwise specified (see 6.2), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.14 in accordance with Table II of MIL-STD-461.

* 4.6 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B or C as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The milling machines covered by this specification are to be used in the shops and depots of activities in the Department of Defense.

* 6.2 Ordering data. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification

b. Type, class and size machine required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

e. Painting, if different (see 3.3.5).

f. Spacing of T-slots, if different (see 3.4.2).

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g. Tape controlled spindle speeds, if required (see 3.4.7).

h. Dual horsepower spindle motor(s), if required (see 3.4.7).

i. MCU requirements, if different (see 3.4.11 through 3.4.11.26).

j. Preparatory functions required, and assignment of codes, if different (see 3.4.11.3).

k. Miscellaneous functions required, and assignment of codes, if different (see 3.4.11.4).

l. Postprocessor additional characteristics required, and computer with which to be used (see 3.4.11.25).

m. Electrical equipment, if different (see 3.8).

n. Voltage characteristics of machine, if different, and voltage for initial wiring (see 3.8).

o. Type of enclosure for motors, if different (see 3.8.1).

p. Coolant heat exchanger, if required (see 3.9).

q. Standard equipment, if different (see 3.11).

r. Optional equipment required (see 3.12 through 3.12.15).

s. Additional accessories and equipment, as required (see 3.13).

t. Electromagnetic interference control, if different (see 3.14).

u. Type fungus proofing, if required (see 3.15).

v. Lubrication chart or plate, if different (see 3.16).

w. Nameplate, if different (see 3.17).

x. Identification plate, if required (see 3.18).

y. Control element identification, if different (see 3.19).

z. First Article inspection, if different (see 4.2).

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aa. Quality conformance inspection, if different (see 4.3).

bb. Electromagnetic interference control test requirements, if different (see 4.5.4 and 6.4).

cc. Level of preservation-packaging required (see 5.1).

dd. Level of packing required (see 5.2).

ee. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, postprocessor, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.20). Consideration should be given to requiring early delivery of certain contract data such as postprocessor, foundation plans, and wiring diagrams.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should include complete information concerning examination, testing, and approval.

* 6.4 Army and Air Force procurements. When equipment purchased for the Department of the Army or the Air Force requires electromagnetic interference control testing, such testing should be specified to be in accordance with MIL-STD-461, Class IIB equipment, except only Test Methods CE03 and RE02 should be required.

* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the Standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - None
Navy - SH
Air Force - None

User Activities:
Army - WC
Navy - AS, OS, MC
Air Force - None

Preparing Activity:
DSA-IP

Project Number:
3417-0104

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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